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REVIEW

Trends in agricultural technology: a review of US patents

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Abstract

Background In recent decades, technological advances have significantly transformed various sectors of the economy, including agriculture. Precision agriculture, as a multidisciplinary and integrative approach, relies on the convergence of diverse technologies to optimize decision-making and enhance productivity, efficiency, and sustainability. In these evolving and complex technological landscapes, patents play a pivotal role, both as a precursor to technological advances and as a critical infrastructure for protecting and advancing innovation progress. The aim of this review is to analyze the evolution of all agricultural technologies, with a focus on precision agriculture, and to compare the position and evolution of each category over time.

Methods We review US patents granted from 2014 to 2024 to identify trends in agricultural technology, with a particular focus on their relevance to precision agriculture. Using a framework adapted from Moreno et al. (2024), we classify patents into eight agricultural technology categories: Automation, Control & Robotics; Bioengineering & Biotechnology; Computing & Cloud Technology; Data Acquisition & Communication; Data Science & Artificial Intelligence; Information Systems; Manufacturing Technologies & Equipment; and Resource-related Technologies.

Results The prevailing categories were Biotechnology & Bioengineering, Manufacturing Technologies & Equipment, and Automation, Control & Robotics. The first two outrank Automation, Control & Robotics in terms of number of patents, but their number of patents declined in the last year, while Automation, Control & Robotics shows a slight but steady growth. A relatively low frequency of patent submission in Data Acquisition & Communication, Data Science & Artificial Intelligence, Computing & Cloud Technology, and Information Systems, underscores the untapped potential for digital transformation in precision agriculture. These technologies are essential for real-time monitoring, advanced data analytics and decision support systems, and are fundamental to precision agriculture. However, it is worth noting that Data Science & Artificial Intelligence has experienced steady growth over the past four years.

Conclusions The interrelated nature of these technology categories highlights the need for an integrated approach to innovation, where advances in one area can synergistically enhance others. This study provides valuable insights for policy makers, researchers and

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industry stakeholders, and highlights the critical role of precision agriculture in addressing global agricultural challenges.

Keywords Innovation · Agricultural technology categories · Patents · Precision agriculture · Emerging trends

Introduction

In recent decades, technological advances have significantly transformed various sectors of the economy including electronics, information and communication technologies, new materials, and modeling (Schwab, 2017). The agricultural sector, driven by the need to increase productivity, reduce environmental impacts, and adapt to climate change, has adopted many technological innovations (Hrustek, 2020; Dayıoğlu et al., 2021; Balyan et al., 2024), leading to the emergence of Precision Agriculture. Precision agriculture, as defined by the International Society of Precision Agriculture (ISPA, 2019), is an integrative approach that relies on the collection, processing, and analysis of data at various spatial and temporal levels to optimize decision-making in agricultural production.

The integration of technologies such as automation, artificial intelligence, and data science has reshaped agriculture, advancing concepts like digital agriculture, climate-smart agriculture, and Agriculture 4.0 (Shepherd et al., 2020; Bertoglio et al., 2021). These innovations are central to the "Third Green Revolution," which emphasizes sustainability and efficiency through technological integration (CEMA, 2018). Previous studies focused on specific aspects of the evolution of agricultural technology. Research on precision agriculture in the United States (US) was categorized into management, monitoring, and other areas (Junior et al., 2024). Abbasi et al. (2022) via execution of a systematic literature review on Agriculture 4.0 highlighted the significant advancements in digital technologies such as autonomous robotic systems, Internet of Things (IoT), and machine learning, and point out that most use cases remain in the prototype phase. Tey et al. (2024) reported that priority innovation themes have demonstrated notable dynamism, evolving through distinct phases. The same authors presented an evolution of agricultural mechanization delineated by four distinct phases: "motorized mechanization" (1960–1999), "mechanical automation" (2000–2009), "digital mechanization" (2010–2019), and "digital automation" (2020–2021).

This evolution underscores the continuous transformation of the sector and the shifting focus of innovation over time. Following this rationale, patents are precursors to innovation (Higham et al., 2021). In modern agricultural, patents play a critical role for generating technological advances, protecting innovation, and driving progress. Patent data can be a valuable resource for analyzing innovation, predicting technology trajectories, and identifying emerging trends and technology gaps (Ernst, 1997; Abbas et al., 2014; Gao & Zhang, 2022). Previous review studies focused on patent synthesis have been conducted to assess technological advances in agriculture related to climate change (Auci et al., 2021), intellectual property (Ren et al., 2017), and biotechnology (Taylor & Cayford, 2003).

Historically, the United States (US) has pioneered technologies tailored to the demands of large-scale production systems. The US remains at the forefront of global agricultural innovation, accounting for nearly 30% of all agricultural patents worldwide (WIPO, 2024). As global agricultural practices increasingly shift toward more centralized and expansive operations, innovations originating in the US are poised to play a pivotal role in improving both the efficiency and sustainability of these evolving farming structures. Innovations have a large impact on global food security (Gouvea et al., 2022). For example, the introduction of patent protections led to more technological progress on plant breeding, increasing crop productivity (Moscona, 2021). Altogether, technological innovations are key drivers influencing farm consolidation for both mid-and large-size farmers in the US (Langemeier & Boehlje, 2017). Technological progress in plant breeding, utilization of fertilizers, improved mechanization, and use of inputs have revolutionized US agriculture, increasing economy of scale and size of farms (Dimitri et al., 2005).

Several studies have attempted to classify agricultural patents and technologies in diverse ways. For instance, Clancy et al. (2020) conducted a study on synthesized U.S. agricultural patents by categorizing them into six subsectors but notably excluded information technology applied to agriculture. In contrast, Balafoutis et al. (2017) proposed a classification framework for precision agriculture, by categorizing technologies into data acquisition, analysis and evaluation, and precision application, but with limited scope and applicability across the agricultural supply chain. Other studies have focused on specific technological classifications. For example, Rovira-Más (2010) classified agricultural machinery sensors into four types: global positioning, feedback control and vehicle position, non-visual monitoring, and local perception. Similarly, Kamran et al. (2016) classified agricultural information systems into four groups based on the services provided to farmers: information, decision, expert, and management support systems. Meanwhile, Lynda et al. (2023) classified an agricultural dataset of Internet of Things (IoT) into six subdomains or applications, different sensors: crop management, irrigation, fertigation management, greenhouse management, plant diseases, and climate. However, they have not taken a broader approach to assess the evolution and comparison of the key technological components that shape agricultural technology. By examining the full spectrum of these technologies, this study provides a comprehensive comparative analysis, evaluating their development and positioning over time. To achieve this, we review US patents from 2014 to 2024 to identify trends in agricultural technology, with a particular focus on their relevance to precision agriculture. Using the classification framework proposed by Moreno et al. (2024), we examine eight key technological categories: (1) Automation, Control, and Robotics, (2) Biotechnology and Bioengineering, (3) Computing and Cloud Technology, (4) Data Acquisition and Communication, (5) Data Science and Artificial Intelligence, (6) Information Systems, (7) Manufacturing Technologies and Equipment, and (8) Resource-related Technologies. These categories reflect the diversity of technological innovations in agriculture related to precision agriculture. Data Acquisition & Communication facilitates the collection and transmission of data, while Computing & Cloud Technology provides the necessary infrastructure for efficient storage and processing. Advanced data analysis, crucial for identifying patterns and trends, is enabled by Data Science & Artificial Intelligence, and its integration into management systems is achieved through Information Systems. Automation, Control & Robotics drives process automation and operational optimization, enabling precise and efficient operations. Manufacturing Technologies & Equipment also plays a critical role in precision agriculture through the development of advanced technologies and equipment. Bioengineering & Biotechnology contribute to precision agriculture by developing crop varieties with specific traits, for example. Resource-related Technologies encompass innovations that focus on optimizing the use and management of critical agricultural inputs such as water, soil, energy, and environmental conditions. The integration of all agricultural technologies allows us to have a comprehensive understanding of the current state of innovation and its potential to shape the future of precision agriculture. This comprehensive analysis of agricultural technology patents provides technology developers and stakeholders in the agricultural sector with a clear roadmap of innovation hotspots and potential areas for investment. Furthermore, the comparative assessment of technological positioning over time reveals gaps and opportunities for interdisciplinary integration, that could accelerate the development of novel solutions that combine previously disparate technologies.

Materials and methods

Collection of patent data

The workflow of this review was structured as shown in Fig. 1. First, the patent search was conducted using the Lens.Org platform (Lens.Org, 2024). The search query filtered records classified under the Cooperative Patent Classification (CPC) (Montecchi et al., 2013) category A01 "Agriculture, Forestry, Animal Husbandry, Hunting, Trapping, Fishing". Then, the search was further refined using the following criteria: (i) granted date between 01-01−2014 and 12–14−2024; (ii) availability of a title; (iii) jurisdiction limited to the United States (This restriction was applied due to the large volume of patents, allowing for a more focused and in-depth analysis within a single jurisdiction); (iv) patents with an Active Legal Status; and (v) document type restricted to Granted Patents. This search resulted in 54,671 patents. As the CPC classifications follow a hierarchical structure in which the order reflects the relative importance of the categories assigned, we selected only those patents in which A01 was the first or second category. This approach ensures that the dataset prioritizes patents with a core agricultural focus while filtering out those where agriculture is a more peripheral aspect. Duplicate patents and patents not related to agriculture were excluded in a further filtering step. The CPC classification system was employed as an initial filter to exclude patents outside the scope of this study, particularly those focused on animal-related technologies. While the search encompassed all subcategories from A01 to A01P within "Agriculture, Forestry, Animal Husbandry, Hunting, Trapping, Fishing," patents classified under A01 K ("Animal Husbandry; Aviculture; Apiculture; Pisciculture; Fishing; Rearing or Breeding Animals, Not Otherwise Provided For; New Breeds of Animals") and A01L ("Shoeing of Animals") were specifically categorized as animal-related and subsequently excluded from further analysis. Then, keyword-based filtering was applied to the titles, using specific terms such as "automat" to capture derived words (e.g., "automatic", "automated") for the Automation and Control Systems category. Through this process, a total of 456 patents were classified into this category. Finally, patents not captured by the previous filters underwent a manual review of titles and abstracts. Each patent was assigned to one of the eight categories defined by the functional taxonomy of agricultural technologies proposed by Moreno et al. (2024).

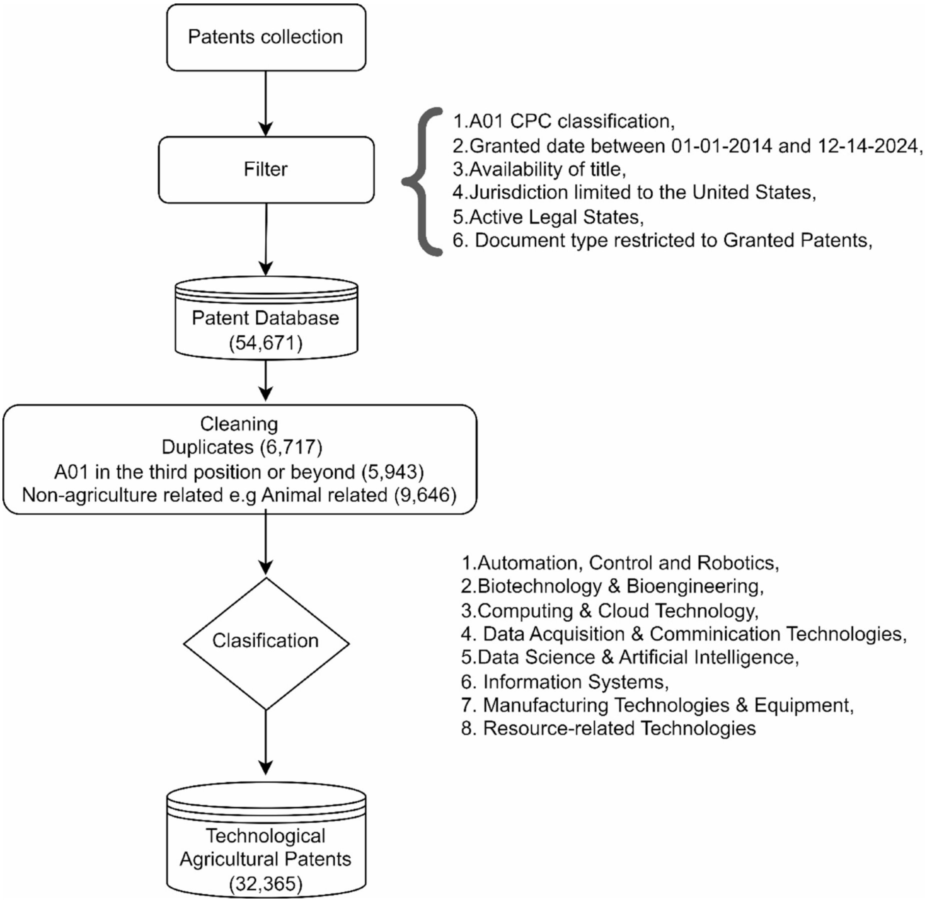

Fig. 1 Workflow of the patent analysis process

Patent classification

The classification of patents followed the framework proposed by Moreno et al. (2024), where each patent was manually assigned to one of eight distinct categories. The categories were:

1. *Automation, Control, and Robotics*: Includes technologies for monitoring and controlling production and services, automatic control systems, and robotics for performing specific tasks with high speed and precision.
2. *Bioengineering and Biotechnology*: Encompasses the integration of natural sciences and engineering applied to organisms, cells, and their components to develop innovative products and services.
3. *Computing & Cloud Technology*: Includes tools and applications for data storage and processing on remote servers, encompassing networks, databases, and software.

4. *Data Acquisition & Communication*: Covers processes related to data collection, processing, storage, and transmission, including image acquisition, remote sensing, sensor technologies, and data transmission.
5. *Data Science & Artificial Intelligence*: Involves tools for deep data analysis using scientific methods and artificial intelligence algorithms.
6. *Information Systems*: Includes organized systems of interrelated components (hardware, software, networks, and people) that collect, process, store, and distribute information to support decision-making and management.
7. *Manufacturing Technologies & Equipment*: Refers to advanced technologies and equipment used in agricultural and manufacturing processes, including actuators, advanced materials, and nanotechnology.
8. *Resource-related Technologies*: Includes innovations focused on the management and optimization of natural resources, covering smart agricultural buildings, carbon capture, renewable energy, soil and environmental conservation, and efficient water and waste management.

These eight categories related to precision agriculture are interrelated, and a patent may cover more than one of them (see Fig. 2).

To validate the accuracy of the manual classification, a Random Forest model was trained using the data obtained from the manual classification to corroborate the precision of the classification (Annex 1).

Network visualization of patent categories

To explore the relationships between patent categories, text analysis and network visualization techniques were combined using the igraph package (Csardi et al., 2024). For this purpose, the text data from patent titles and abstracts were first preprocessed, including the removal of stop words and the application of lemmatization to clean the dataset. The co-occurrence of keywords across different categories was then calculated using the widyr package (Robinson, 2022). A document-term matrix (DTM) was subsequently constructed to capture keyword frequencies, and a similarity matrix was derived using correlation metrics to quantify the connections between categories based on shared keywords. All text processing and network analysis were performed using R software (R Core Team, 2023). This approach provides a visual and quantitative representation of how different technologies interact within the broader agricultural innovation ecosystem.

Results

Agricultural technology categories and distribution for US patents

The total number of patents linked to agricultural technology from 2014 to 2024 was 32,365 (Table 1). The terms identified within each category enable us to define and differentiate them. However, it is noteworthy that certain terms, such as "control" and "sensors" are present in multiple categories (Table 1). The most common category of US patents filed over the last decade was biotechnology and bioengineering (52.4%), followed by manufacturing

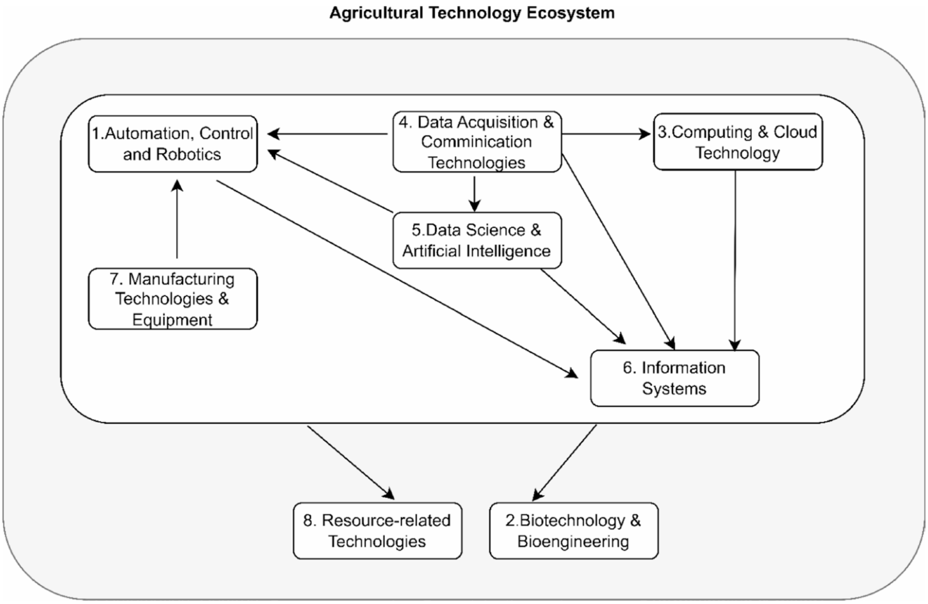

Fig. 2 Agricultural technology ecosystem. 'Automation, Control, and Robotics' is intricately linked to 'Data Acquisition & Communication' and 'Data Science & Artificial Intelligence'. For example, in the patent *'System and method for controlling an agricultural system based on soil analysis'*, a controller is configured to receive signals from an agricultural soil analyzer. Although generating control action often relies on data acquired through these categories, patents of this nature were classified under 'Automation, Control, and Robotics.' Additionally, 'Automation, Control, and Robotics' is also closely related to 'Manufacturing Technologies and Equipment'. For example, in the patent *'Combine harvester having a driver assistance system'*, actuators are integrated with automated functions to enhance operational efficiency. While such innovations involve manufacturing components, their primary focus on automation and control places them within the 'Automation, Control, and Robotics' category. The category 'Information Systems' is related to 'Automation, Control, and Robotics', 'Data Acquisition & Communication', 'Data Science & Artificial Intelligence' and 'Computing & Cloud Technology'. For example, in this patent, *Visual Information System and Computer Mobility Application for Field Personnel*, the focus is on the collection and visualization of crop data, combined with communication technologies and information processing. Although this involves aspects of automation, communication, and data science, the patent was classified as 'Information Systems'. 'Data Acquisition & Communication' is closely related to 'Data Science & Artificial Intelligence', as data collection serves as the basis for deep data analysis tools using scientific methods and artificial intelligence algorithms. In addition, 'Data acquisition and communication' is interconnected with 'Computing & Cloud Technology', when data is often stored and processed in the cloud. Broadly, all these technologies are applied to agriculture, which encompasses two fundamental aspects: crop production, closely linked to 'Biotechnology & Bioengineering,' and environmental and infrastructure management, which is closely tied to 'Resource-related Technologies.'

technologies and equipment (35.8%) and automation, control and robotics (8.2%). In contrast, the less common categories, mainly based in the field of agriculture, included Computing & Cloud Technology (0.06%) and Information Systems (0.27%) (Table 1).

Agricultural technology network

Most of the technologies were strongly connected to at least one other category (Fig. 3). Robust connections between categories indicate areas of research that may be closely

Table 1 Patent categories, frequent terms included, and total number by category

Categories	Frequent terms included in patents	Number (%)
1. Automation, Control, and Robotics	Control, Configured, Vehicle, Controller, Sensor, Position	2551 (7.9)
2. Biotechnology & Bioengineering	Plant, Variety, Soybean, Invention, Maize, Producing	17,055 (52.7)
3. Computing & Cloud Technology	Irrigation, Data, Information, Software, Sensor, Management	20 (0.06)
4. Data Acquisition & Communication	Sensor, Data, Configured, Soil, Control, Seed	415 (1.3)
5. Data Science & Artificial Intelligence	Crop, Image, Machine, Sensor, Control, Soil, Model	511 (1.3)
6. Information Systems	Field, Information, User, Crop, Plant, Irrigation	89 (0.27)
7. Manufacturing Technologies & Equipment	Assembly, Device, Frame, Configured, Position, Material	11,562 (35.7)
8. Resource-related Technologies	Water, Soil, Plant, Air, Greenhouse, Solar	162 (0.5)

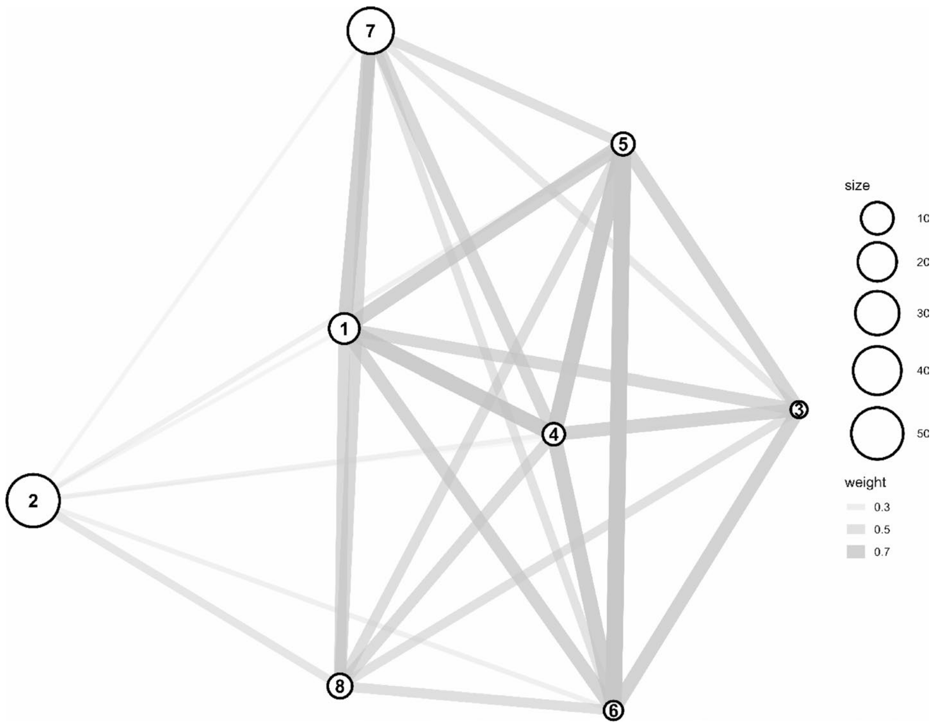

Fig. 3 Network of relationships between agricultural technology categories. The number inside the bubbles corresponds to the patent categories (1) Automation, Control, and Robotics, (2) Biotechnology and Bioengineering, (3) Computing and Cloud Technology, (4) Data Acquisition and Communication, (5) Data Science and Artificial Intelligence, (6) Information Systems, (7) Manufacturing Technologies and Equipment, and (8) Resource-related Technologies. Bubble sizes indicate the proportion of patents in each category, and the width of the connectors indicate the strength of the relationship between the categories

related, while weaker connections suggest more peripheral relationships. For instance, Data Acquisition & Communication was highly interconnected with the rest, particularly with Automation, Control and Robotics, Computing & Cloud Technology, Data Science & Artificial Intelligence, and Information Systems, highlighting its central role in the overall research framework. Other strong relationships were Data Science & Artificial Intelligence also had strong links with Information Systems, Data Acquisition & Communication and Automation, Control and Robotics. On the other hand, Biotechnology & Bioengineering had weaker links, suggesting that it was more peripheral (Fig. 3).

Evolution of agricultural US patents

Over the 10 years analyzed, Biotechnology & Bioengineering was the category with the highest number of patents; however, this number showed a decline from 2021 to 2024 (Fig. 4A). 'Manufacturing Technologies & Equipment' showed an increase from 2014 to 2019, followed by stabilization from 2019 onwards (Fig. 4A). On the other hand, Automation, Control and Robotics showed a slight growth with an increase of 213% over the last 10 years relative to the initial level (Fig. 4A).

The initial gap a decade ago between Biotechnology & Bioengineering and Manufacturing Technologies & Equipment was approximately 948 patents in favor of Biotechnology & Bioengineering. Over time, the gap has gradually narrowed to just 58 patents in the last year (Fig. 4A). The categories with the lowest share of patents showed the highest growth over the last decade (Data Science & Artificial Intelligence 641.7%, Data Acquisition & Communication 210.5%). Conversely, Computing & Cloud Technology, Information Systems and Resource-related Technologies remained relatively stagnant. It is worth noting that Data Science & Artificial Intelligence was the category with the highest growth from 2020 onwards (Fig. 4B).

Discussion

This review of US agricultural patents reflects the current state of agricultural technology innovation and shows how categories related to precision agriculture have evolved. These findings highlight areas of significant progress and identify emerging trends that may shape the future of agriculture.

The US agricultural biotechnology and bioengineering have been closely linked to the strengthening of intellectual property rights, which have driven exponential growth in these sectors since the 1980 s (Buccola & Xia, 2004). The manufacturing technologies and equipment category, which has seen the largest growth in patents over this period, began to develop in the 1960 s with incremental innovations aimed at motorizing heavy machinery to increase agricultural yields, and mechanical automation became increasingly important in the 2000 s (Tey et al., 2024). Similarly, as reported in our study, Mann and Püttmann (2018) reported an upward trend in automation patents from 1976 to 2014. In addition, Sozzi et al. (2018) found a significant increase in data and automation patents, from 5% in 2010 to 15% in 2018.

For the low frequency categories, such as Data Acquisition & Communication (Kurfees et al., 2020), Data Science & Artificial Intelligence (Nageye et al., 2024), Computing &

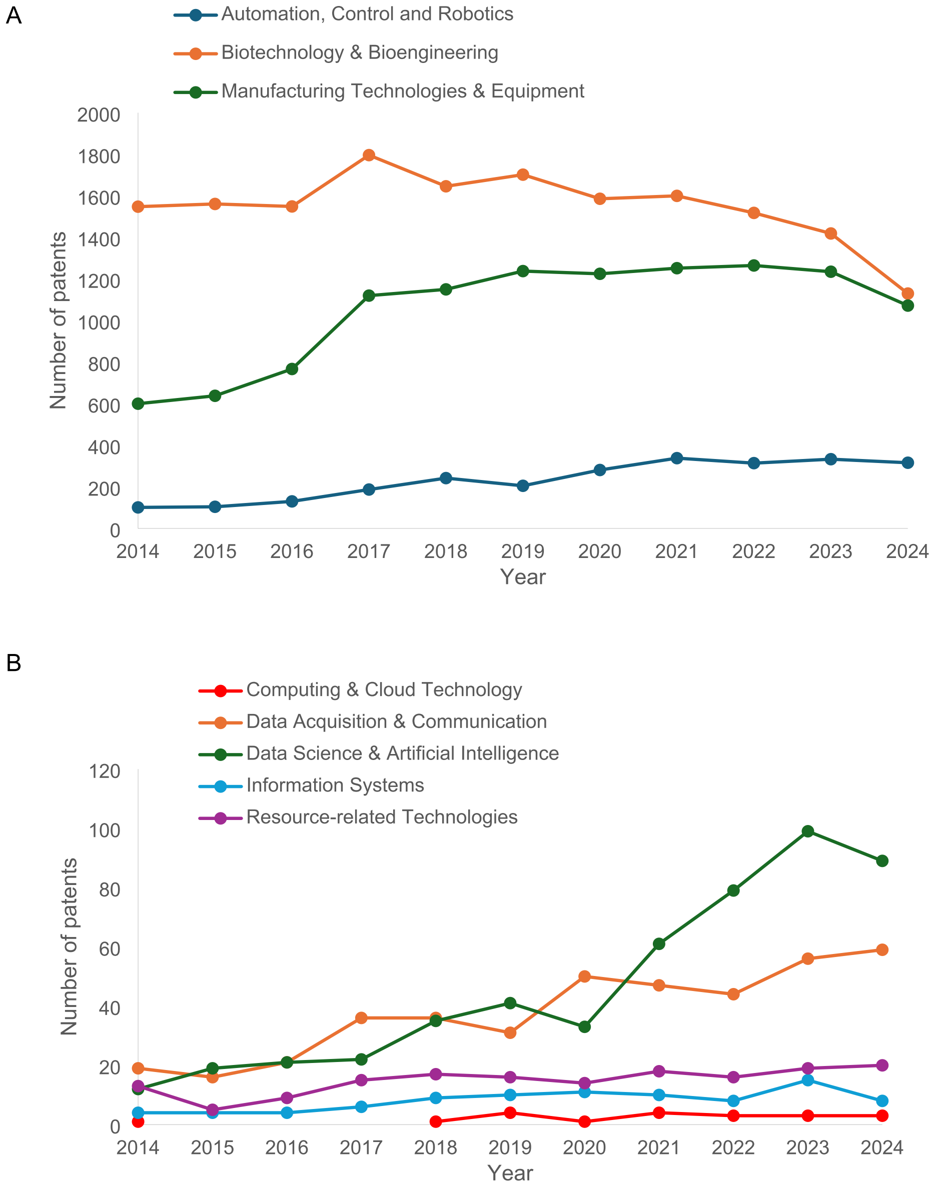

Fig. 4 Evolution of agricultural patents in the United States. Panel A presents the categories with the highest number, while Panel B illustrates those with the lowest

Cloud Technology (Bataev et al., 2018), Information Systems (Vassilakopoulou et al., 2023), and Resource-Related Technologies (Gorjian et al., 2022), significant progress has been made, but this is not reflected when focusing only on agriculture. Thus, it is hypothesized that many of these innovations are not deliberately targeted at agriculture as a key sector, but that the deployment of these innovations occurs as other applications of these technologies are explored in different sectors. For example, the integration of data science and artificial intelligence has profoundly transformed the healthcare sector (Gruson et al., 2020). For agriculture, recent review papers have highlighted the contribution of this technology category to advancing food security (e.g., Wolfert et al., 2017; Patricio and Rieder, 2018; Liakos et al., 2018; Benos et al., 2021).

If current trends continue, it is expected that categories related to precision agriculture, such as Automation, Control and Robotics, Data Acquisition & Communication, and Data Science & Artificial Intelligence, will continue to experience significant growth in the coming years. These changes increase productivity, strengthen food security by lowering prices, but can also slightly increase greenhouse gas (GHG) emissions due to increased production for export, displacement of labor, and require significant investment (Bazargani et al. 2024; Wang et al., 2024). Therefore, it is crucial that the adoption of these technologies leads to a sustainable intensification, ensuring that productivity gains do not come at the expense of environmental or social well-being.

The eight categories proposed by Moreno et al. (2024) offer a comprehensive framework for classifying agricultural patents. Our analysis highlights the inherently interconnected nature of agricultural technologies. The terminological overlap between categories underscores that, while the functional taxonomy enables categorization based on the primary focus of each patent, the boundaries between categories remain permeable, with concepts flowing across technological domains. This finding suggests that the future of agricultural innovation will likely depend not only on advancements within individual categories, but also on the strategic integration of complementary technological domains (Pigford et al., 2018). For instance, Massabni and da Silva (2019) demonstrate how emerging technologies such as artificial intelligence, robotics, and 3D printing play a critical role not only as innovations but also in advancing fields such as biotechnology. Jin and Han (2024) highlight the vast potential of robotic arms in precision agriculture. Bouali et al. (2021) demonstrate that the integration of technologies such as smart water meters (data acquisition and communication), renewable energy systems (resource-related technologies), and smart irrigation (automation, control, and robotics) not only improves resource efficiency but also plays a key role in advancing sustainable practices. This synergy between different technologies highlights the need for a holistic approach to agricultural research and development. A few opportunities to be highlighted as future research are (i) quantitatively assessing the quality of patents, this point is out of scope for the current study but it is a clear aspect that deserves further attention (Higham et al., 2021); (ii) impact is related to disruptiveness, a recent study published by Park et al. (2023) analyzed a large dataset across disciplines, highlighting that patents are becoming less disruptive and suggesting focusing on quality for innovations (Bhattacharya & Packalen, 2020); (iii) development of a more systematic approach to classify patents, potentially in a near future guided by AI, patent classification could be less challenging but a better classification system is a critical point that also needs to be discussed. As related to this last point, we found, although in a relatively small proportion (< 0.001% of the total), non-agriculture-related patents classified under CPC codes such as A01. Automated classification based on keywords is complicated due to the overlap of common terminology across different categories, including terms like “data,” “control,” and “device.” A single patent often covers multiple technologies, requiring careful analysis to clearly identify the overall primary focus of each patent. In addition, the vague language used to write patents can also limit the ability to identify other useful innovations related to agriculture if relevant terms are not disclosed (Arinas, 2012). Finally, relying solely on patent data may not fully capture the extent of technological innovations, especially those protected by other forms of intellectual property. Future studies could focus on integrating a patent analysis with a bibliometric study to improve the overall picture of the past and the state of progress of agricultural technologies.

In this study, we prioritized the functional and technical dimensions of technology classification, following the framework proposed by Moreno et al. (2024). However, technologies can also be classified along several other dimensions, including adoption stage, maturity level, business model impact, reusability, anticipated development trajectory, and overall effectiveness. Future research could enhance this classification approach by integrating additional perspectives, such as technology readiness levels (TRLs) or market adoption rates, to provide a more comprehensive and multidimensional understanding of technological evolution within the agricultural sector.

The findings of this study provide valuable implications for policy makers and agricultural extension agencies. The predominance of patents in Biotechnology and Bioengineering and Manufacturing Technologies and Equipment, along with the steady growth in Automation, Control, and Robotics, highlights key areas where innovation is currently concentrated. However, the relatively low frequency of patents in digital categories (i.e., Data Acquisition and Communication, Data Science and Artificial Intelligence, Computing and Cloud Technology, and Information Systems) suggests untapped potential that warrants greater policy and educational attention. Policymakers should consider targeted incentive programs to stimulate innovation in these underrepresented but critical areas for the advancement of precision agriculture. Agricultural extension agencies, in turn, could focus their efforts on developing educational programs that facilitate the adoption of emerging digital technologies, especially in light of the rapid growth observed in Data Science and Artificial Intelligence. Moreover, the interconnected nature of technological categories—evident in our network analysis—suggests that the most effective innovation policies will be those that promote integrated, multidisciplinary approaches rather than treating each technological domain in isolation. This could be reflected in extension programs that not only teach about individual technologies but also emphasize how their integration can maximize benefits for agricultural productivity and environmental sustainability.

Conclusion

This study highlights the transformative potential of technological innovation in shaping the future of precision agriculture. Advances in automation and manufacturing technologies have been key drivers of the current technological revolution, while lower patent frequencies in digital technologies highlight areas for future growth. The interrelated nature of these technologies underlines the importance of an integrated approach to innovation, where advances in one category can synergistically enhance others.

As the agricultural sector continues to evolve, precision agriculture will play a critical role in addressing global challenges such as food security, resource efficiency, and sustainability. By exploiting the convergence of different technologies, precision agriculture can optimize decision-making, increase productivity and promote sustainable practices. Efforts will continue to monitor these trends and adapt to the changing technological landscape to ensure that agriculture can meet the challenges of the future.

Annex 1: validation of patent classification

To validate the accuracy of the manual classification, a text-mining and machine-learning approach was implemented. Patent titles and abstracts were tokenized, lemmatized, and filtered to remove stop words, resulting in a clean dataset. A Document Term Matrix (DTM) was then created by counting the occurrences of each term across documents. This DTM served as input for a Random Forest model (Breiman, 2001) using the randomForest package (Liaw & Wiener, 2002).

The dataset was split into training and testing subsets using an 80/20 ratio. Model training was performed using the training subset, and model validation was conducted on the testing subset. Model validation was performed by comparing the trained model's predictions with the manually assigned categories in a confusion matrix, which cross-tabulates observed and predicted classes. Key metrics such as accuracy, 95% confidence intervals, No Information Rate (NIR), p-value, Cohen's kappa (Cohen, 1960), and sensitivities were reported (Fig. 5). All statistical analyses were conducted using R software (R Core Team, 2023). To ensure optimal model performance, it is essential to attain a high level of accuracy, ideally exceeding 0.8 or 80%. Additionally, narrow confidence intervals surrounding key metrics are crucial to guarantee precise estimates with minimal variability. The model's accuracy should be markedly superior to the No Information Rate (NIR) to substantiate its superior performance compared to random guessing based on the most frequent class. A p-value <0.05 indicates that the model's performance is significantly superior to that which could be expected by chance.

The results of the model evaluation showed that the accuracy was high, and the Cohen's Kappa value was 0.43, indicating a moderate level of agreement between the model's predictions and the actual classifications, which exceeds what would be expected by chance. A

	Observed							
Predicted	1	2	3	4	5	6	7	8
1	40	23	0	0	3	0	36	1
2	134	2708	2	23	18	5	547	39
3	0	0	0	0	0	0	0	0
4	3	0	0	1	0	0	1	0
5	2	0	0	0	0	0	1	0
6	0	0	0	0	0	0	1	0
7	314	675	2	58	81	11	11647	123
8	3	1	0	0	0	0	8	1

Overall Statistics

Accuracy: 0.67

95% CI: (0.6637, 0.6866)

No Information Rate: 0.5232

P-Value [Acc > NIR]: <2.2 e -16

Kappa: 0.4258

Fig. 5 1. Automation, Control and Robotics, 2. Biotechnology & Bioengineering, 3. Computing & Cloud Technology, 4. Data Acquisition & Communication, 5. Data Science & Artificial Intelligence, 6. Information Systems, 7. Manufacturing Technologies & Equipment, 8. Resource-related Technologies *p*-value of less than 0.05 suggests that the model's performance is significantly superior to that which could be expected by chance.

Confusion matrix for the model performance evaluation.

Based on the confusion matrix, the model demonstrated varying sensitivities across different categories. The categories with the highest sensitivity were Biotechnology & Bioengineering and Manufacturing Technologies & Equipment, with sensitivities of 0.79 and 0.73, respectively. Automation, Control, and Robotics, as well as Data Science & Artificial Intelligence, exhibited moderate sensitivities of 0.8 and 0.12, respectively. The remaining categories, which appeared less frequently, showed almost negligible sensitivity.

The model achieved an overall accuracy of 0.67%, indicating that it correctly classified approximately 67% of the patents. The Kappa statistics were 0.42, reflecting substantial agreement between the predicted and actual classifications, surpassing what would be expected by chance. The 95% confidence interval for accuracy ranged from 0.663 to 0.686%, providing a robust estimate of the model's performance. The No Information Rate (NIR) (the proportion of categories that fall into the majority class) was 0.52%, representing the accuracy of a naive model that always predicts the most frequent class.

The Random Forest model demonstrated strong predictive ability, particularly for more prevalent categories, there was variability in performance across different categories. Some categories exhibited higher misclassification rates. The substantial Kappa value underscores the model's reliability. Although the results suggest that the model performs well overall, refinement may be necessary to improve accuracy, particularly for underrepresented categories.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11119-025-10257-x.

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Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

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